Pothanaboyena Pavan

AI & Data-Science Student

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Professional Summary

Results-driven Data Science undergraduate specializing in Computer Vision and Deep Learning with proven experience building and deploying production-grade AI models. Adept at developing end-to-end ML pipelines — from data preprocessing to REST API deployment. Seeking a data-focused internship to apply expertise in object detection, transfer learning, and model optimization to solve real-world problems at scale.

Education

S.R.K.R Engineering College, Bhimavaram — Sep 2023 – Apr 2027
(3/4) B.Tech – Artificial Intelligence & Data-Science — CGPA: 9.14/10

Sri Chaitanya Junior College — 2021 – 2023
Intermediate Education — Percentage: 98%

Technical Skills

Programming Languages:
Python, Java, C, SQL

Data Science & ML Libraries:
Python Libraries, TensorFlow, Keras

Web Development:
HTML, CSS

Core Concepts:
Data Structures & Algorithms

Project Experience

Shrimp Count Prediction — Government-Sponsored Project — Apr – Aug 2025

- Trained YOLOv8 object detection model on 8,000+ labeled images, achieving 91% mAP at 24 FPS for accurate real-time shrimp counting in aquaculture pond images.
- Built an end-to-end inference pipeline using OpenCV for image preprocessing, detection, and automated count output with visual bounding-box overlays.

Tech Stack: Python, YOLOv8, OpenCV, TensorFlow, Pandas, NumPy

AI-Powered Task Management Platform — 2025

- Designed a full-stack REST API with Node.js/Express, PostgreSQL & Prisma ORM, implementing JWT auth, role-based access control, audit logging, and hierarchical task/subtask workflow with automatic parent-child status sync.
- Engineered a FastAPI ML microservice with sentence-transformer embeddings and a custom logistic regression intent classifier to auto-generate subtasks with confidence-based fallback to prevent hallucinations.

Tech Stack: Python, Node.js, Express, PostgreSQL, Prisma, FastAPI, JWT, Scikit-learn, Docker

Gastrointestinal Disease Prediction DL Model — Jun 2025 – Jul 2025

- Developed a deep learning pipeline to process and augment a dataset of endoscopic images for disease classification.
- Implemented a transfer learning model using ResNet50 and TensorFlow, achieving 82% prediction accuracy on the validation set.

Tech Stack: Python, TensorFlow, Keras, ResNet50, NumPy, OpenCV

Achievements and Certifications

- Certifications regarding C, Python Programming, Cybersecurity – Cisco Networking Academy (2023)
- Hack n Clash Coding Competition Certificate (2024)
- Participated in Smart India Hackathon (2024)
- Participated in Amaravati Quantum Valley Hackathon (Stock Market Analysis using VQC) (2025)

Leadership & Activities

- Coordinated logistics for a college event attended by over 1,200 students as a Club Member of ISTE.

Interests

Algorithmic Problem-Solving, Machine Learning Research, Technology Trends, Being Aware Of Current Affairs.